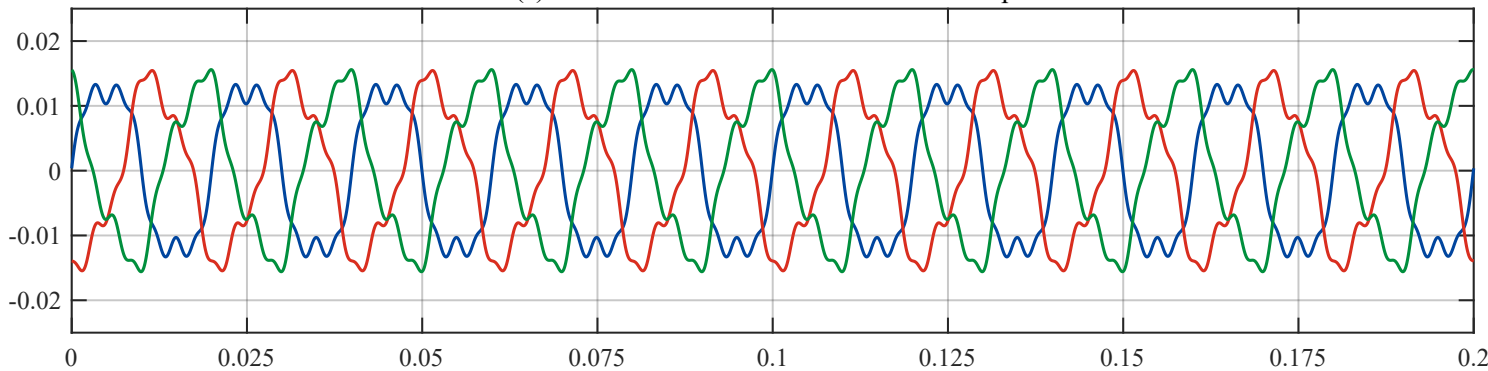
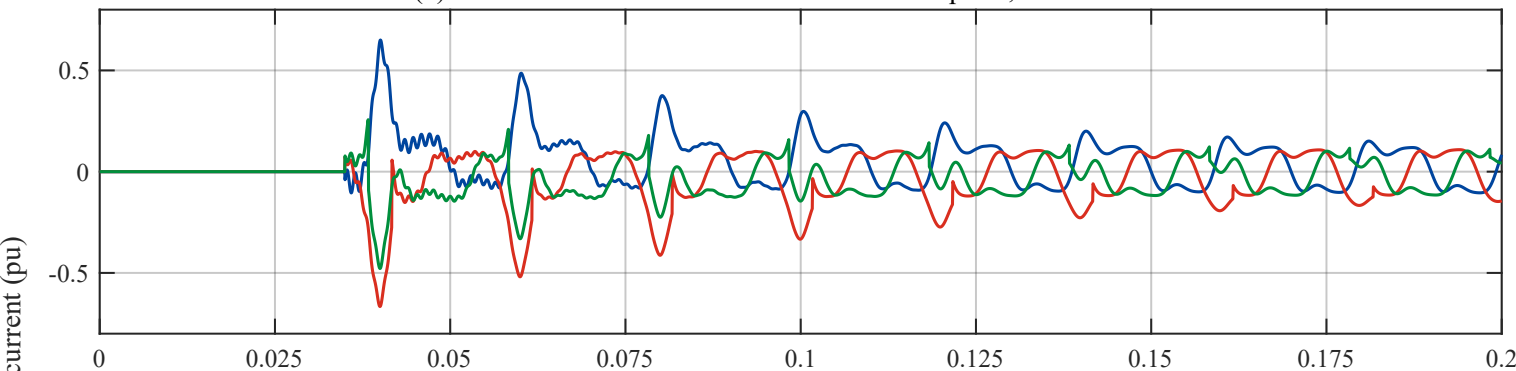


Phase A Phase B Phase C

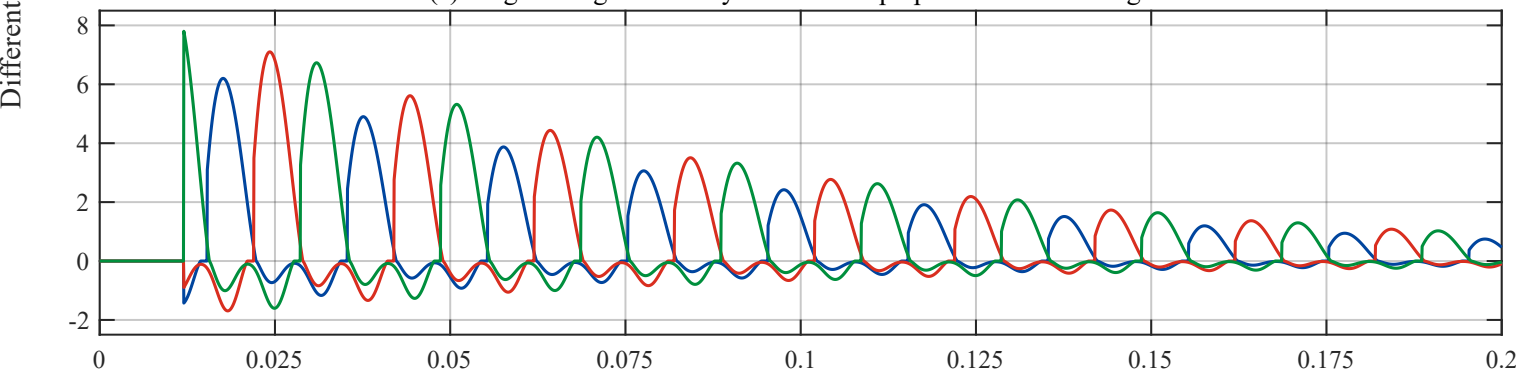
(a) Normal load: differential current < 0.02 pu



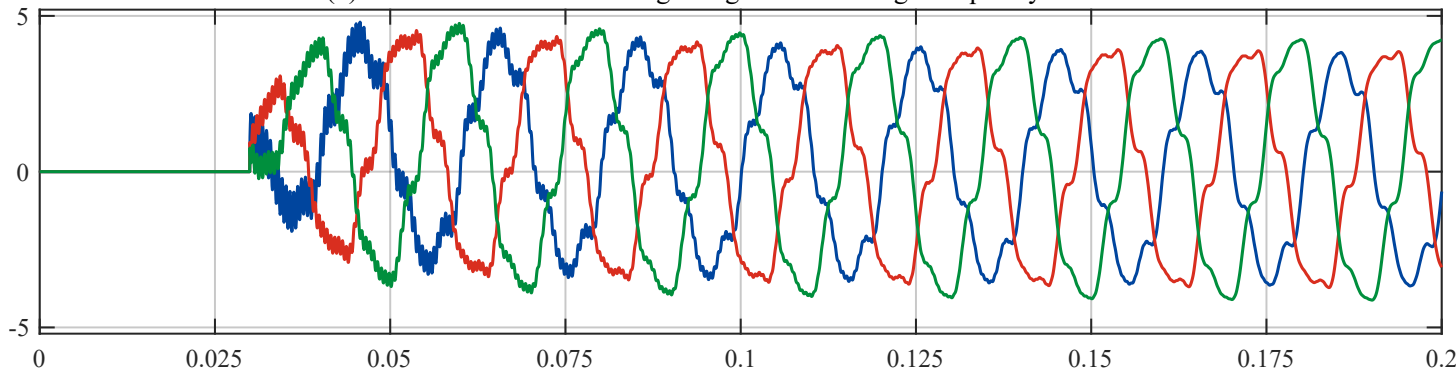
(b) External fault with CT saturation: transient spikes, THD $\approx 34.7\%$



(c) Magnetizing inrush: asymmetric 5-8 pu peaks with dead angles



(d) Internal fault: sustained high magnitude with high-frequency transients



Time (s)